

OligoTox-Kidney — Methodology Document

NIH/NCATS OligoTox Open Data Challenge, Phase 2 · Endpoint: kidney toxicity

Phase 2 methodology deliverable. Target length ≤ 5 pages when rendered to PDF. The long-form internal documentation is METHODOLOGY.md; this is the submission version.

1. Nature of the dataset, and what "materials and methods" means here

OligoTox-Kidney is an **in-silico curation** of pre-existing, publicly reported nephrotoxicity data. No wet-lab experiment was performed. The materials are therefore **documents**, and the methods are **source identification, extraction, harmonisation, grading, provenance capture and quality control** — i.e. how the dataset was assembled and computationally processed.

We state this plainly at the top because it determines what can and cannot be reported below. In particular it is why §5 reports oligo *identity* characterisation in full and oligo *purity* as absent.

Scope control. The dataset is strict-kidney. This is verified rather than asserted: all 246 rows carry `is_kidney_specific = TRUE`; `tissue` takes only kidney (79) and proximal_tubule (167); and an automated audit confirms zero hepatic readouts, zero hepatic system models and zero hepatotoxicity sources cited. Hepatotoxicity source material held elsewhere in the repository contributes **no rows** to this dataset.

2. Materials — sources

Sources were prioritised kidney-first and catalogued in SOURCES.md with stable identifiers, redistribution status and acquisition state.

ID	Material	Contribution
N3	US Patent 11,105,794, <i>In vitro</i> nephrotoxicity screening assay	69 rows — Table 1 (rat in vivo, 21 compounds), Table 2 (human PTEC / PTEC-TERT1 EGF panel)
N4	US Patent 11,479,818, companion patent (EGFR/KIM-1 readout)	81 rows — Table 5 (rat primary PTEC, EGFR mRNA + KIM-1 mRNA/protein)
WS	FDA/EMA labels and trial publications, search-derived	36 rows (flagged; see §6.2)
M1	Moisan 2017 (PMC5363415)	11 rows — human PTEC-TERT1 EGF/ATP panel
N2	Janssen 2019 (PMC6796739)	10 rows — drisapersen, patients + monkey + human ciPTEC
K1	Sandelius 2020 (PMID 33084520)	9 rows — urinary kidney biomarker panel
A1–A13	FDA/EMA labels, prescribing information, pivotal trial and case-report literature	21 rows — marketed-drug clinical anchors
REV	Wu 2022 (PMC10174585)	4 rows — marketed-ASO renal review, used for cross-checking

Patent material is public domain and may be reproduced. Journal and label material carries per-row redistribution status (`summary_stat`, `derived_features_only`, `verify`).

3. Methods — data model

Two normalised UTF-8 CSV tables joined on `oligo_id`:

- `oligos.csv` (65 × 20) — one row per unique oligonucleotide: identity, sequence, chemistry and design predictors.
- `measurements.csv` (246 × 25) — one row per **oligo × system/subject × delivery × concentration/dose × readout**. One oligo at one concentration read on two biomarkers is two rows.

A denormalised analysis view (`oligotox_kidney_merged.csv`, 246 × 46) is **generated by script and never hand-edited**; the two normalised tables remain the source of truth. A derived `human_animal_bridge.csv` (15 oligos) materialises the compounds carrying evidence on both sides of the human/animal divide.

Missing values are the literal string TBD — never guessed, never imputed as zero. The full data dictionary, controlled vocabularies and grading rubric are in `schema.md`.

4. Methods — extraction

Every transformation is a committed, re-runnable script. No derived value is hand-entered.

4.1 Full-text extraction from primary sources

PDFs were parsed with layout-preserving text extraction (`pdfplumber`). Per-measurement values, doses, sequences and exact loci were transcribed into the schema with the source table or figure recorded in `source_table`.

Two hazards required specific countermeasures, because both produce plausible wrong answers rather than obvious failures:

- **Column interleaving.** The patent table pages extract, under naive text extraction, as a flat number stream that cannot be mapped back to (compound, concentration, system, timepoint). Both table extractors therefore parse the *layout-preserving* rendering and **validate parsed cells against anchor values read from the printed table** before writing anything; each refuses to write a partial table if the cell count is wrong.
- **Case-encoded chemistry.** In gapmer sequences, upper case denotes 2'-MOE/cEt wings and lower case the DNA gap. Sequence validators must be case-insensitive; a case-sensitive check reports these correct rows as malformed.

4.2 WHO INN nomenclature derivation

For INN-named oligonucleotides, WHO publishes the complete residue-by-residue chemical nomenclature (2'-O-methyl-P-thiocytidylyl-(3'→5')-...). Sequence is therefore recoverable by **deterministic parse** rather than transcription. This supplied 20 sequences.

Two properties make it safer than reading a sequence off a table, and both were enforced:

- 1 **Direction is explicit and asymmetric.** INN writes one strand with (3'→5') linkages (listed 5'→3') and its partner with (5'→3') linkages (listed 3'→5', hence reversed on output). Mishandling this yields a silently *reversed* strand. The parser was validated by reproducing two sequences already present in the table, character-for-character, before any new row was written.
- 2 **Results are self-checking.** Every duplex was required to show its guide strand as the exact reverse complement of its sense strand (ignoring 3' overhangs), and residue counts were cross-checked against the published molecular formula wherever the phosphorus count fixes the length (e.g. viltolarsen P20 → 21 nt; alicaforsen P19S19 → 20 nt).

4.3 Regulatory label retrieval

FDA prescribing information was read via **DailyMed** structured product labels. Renal content was extracted from the Warnings and Precautions, Adverse Reactions and Use in Specific Populations sections, recording the section number in `source_table`.

5. Methods used to purify and characterise oligo identity

The Phase 2 announcement asks specifically for this. For a curation-type dataset the two halves have different answers, and we report them separately rather than collapsing them.

5.1 Purity — absent, and verified absent

`purity_pct` and `purity_method` are TBD **for all 65 oligonucleotides**.

This was established, not assumed. Both source patents were searched programmatically for purity, HPLC, UPLC, LC-MS and mass-spectrometry language; **neither reports any** (the only "characterised" matches describe chemical structure, not oligo purity). Regulatory labels and trial publications do not publish per-batch analytical purity either.

No wet-lab work was performed, so no purity determination could be made independently. This is a limitation of the curation approach as executed, and was not closed by further extraction. It is recorded as an explicit TBD column with the reason in notes, rather than left as an unexplained blank.

5.2 Identity — established for 55 of 65, with the method recorded per oligo

`identity_confirmation` records **how** each oligo's identity was established:

Method	n	Basis
<code>patent_sequence_listing</code>	25	Formal SEQUENCE LISTING of a US patent, validated base-for-base against the listing rather than the examples table
<code>who_inn_chemical_nomenclature</code>	20	Deterministic parse of WHO INN nomenclature, reverse-complement and molecular-formula checked (§4.2)
<code>regulatory_label</code>	7	Sequence published in an FDA/EMA label
<code>peer_reviewed_publication</code>	3	Sequence printed in a primary publication
<code>not_established</code>	10	Sequence itself is TBD; identity unconfirmed

The 10 unconfirmed are structural, not clerical: 6 proprietary research compounds whose sponsors never published sequences, 2 class-level aggregate rows for which a single sequence is not meaningful, and 2 development-code compounds that never received an INN.

Where a sequence is present, the **location of chemical modifications** is carried by `sugar_modifications` (all 65), `gapmer_design` (wing-gap-wing motif, 40), `ps_count` (phosphorothioate linkage count, 38) and, for gapmers, position-resolved case encoding within `sequence_5to3` itself.

6. Grading, quality control and known limitations

6.1 Grading

Each measurement carries an ordinal `nephrotox_grade` (0–3) assigned from the reported endpoint per the rubric in `schema.md`: 0 no signal; 1 mild/functional/reversible without viability loss; 2 moderate injury-biomarker or histopathology; 3 severe AKI, glomerulonephritis or dose-limiting toxicity.

For the two continuous biomarker panels, grades were thresholded, and **the thresholds are anchored rather than chosen freely**: in each case the floor is set so the source's own designated innocuous reference compound grades 0 on every one of its cells. After application, grade tracks the source's own in-vivo toxicity classification monotonically. Both rubrics are stated in full in the extraction scripts and in `schema.md`.

The two patent panels use **different normalisations** — % of saline versus % of the innocuous reference compound — recorded in `readout_unit` so they are never pooled.

All grades carry `grade_provisional` pending subject-matter sign-off.

6.2 Quality control

Automated checks run after every ingestion round: schema conformance against the `schema.md` enums; referential integrity (`measurements.oligo_id` → `oligos.oligo_id`, 0 orphans); no duplicate primary keys; grades in {0,1,2,3}; and — added after the finding below — **duplex self-consistency**, requiring each siRNA guide strand to be the exact reverse complement of its sense strand.

A corrected error, disclosed. Final QC found 21 rows (19% of the then-dataset) recording a rat study as a mouse study, with a published dose left as TBD. The source patent binds its Table 1 to a method section naming "Wistar Han Crl : WI (Han) male rats", dosing "at 40 mg/kg on days 1 and 8", a "Multiplex MAP **Rat** Kidney Toxicity Magnetic Bead Panel 2", and sacrifice "on day 15". All four statements were verified against the patent before correction. Species distribution moved from mouse 30 / rat 8 to rat 29 / mouse 9.

A provenance/outcome confound, quantified and mitigated. Clinical rows cross-tabulated by sourcing quality against outcome showed provenance predicting outcome almost perfectly — 0 of 20 search-derived rows reached grade ≥2 against 11 of 22 anchor-sourced rows, one-sided Fisher $p = 4.5 \times 10^{-11}$. Direct retrieval of 7 unverified absence claims left only one standing as a measured negative. Two mitigations were applied: three approved drugs with *measured* human negatives were added (moving the association to $p = 1.65 \times 10^{-11}$), and the schema now carries `renal_endpoints_measured`, separating "measured and unremarkable" from "never looked". 13 grade-0 clinical rows are explicitly flagged as not supported as measured negatives. Full analysis in `CLINICAL_VALIDATION.md`.

6.3 Limitations carried into the release

- Grades are provisional pending scientific sign-off.
- Purity is absent for all 65 oligos (§5.1).
- 36 rows remain search-derived; 13 grade-0 clinical rows are flagged unsupported.
- The confound is reduced, not eliminated.
- 39 of 246 rows lack a numeric dose, mostly awaiting FDA Pharmacology/Toxicology reviews.
- The human/animal bridge is 15 oligos — sufficient for descriptive translation analysis, not for a fitted species-correction model.

7. Reproducibility

Every derived artefact is regenerable from the two canonical tables by committed scripts:

<code>scripts/fill_inn_sequences.py</code>	sequences from WHO INN nomenclature
<code>scripts/extract_patent_table2.py</code>	human in-vitro EGF panel (US 11,105,794 Table 2)
<code>scripts/extract_n4_table5.py</code>	rat in-vitro EGFR/KIM-1 (US 11,479,818 Table 5)
<code>scripts/fix_patent_species.py</code>	the rat/mouse correction (§6.2)
<code>scripts/add_identity_characterization.py</code>	identity + purity fields
<code>scripts/add_endpoint_provenance.py</code>	<code>renal_endpoints_measured</code>
<code>scripts/split_human_animal.py</code>	<code>subject_class</code> + bridge view
<code>scripts/build_merged.py</code>	the denormalised analysis view

All source PDFs cited above are included in the repository under `sources/` so that any extracted value can be checked against its origin without re-acquiring the document.